

HDOC Day-25

Generation of Decreasing Patterns Using a Menu-Driven C Program

Aim

To write a menu-driven C program using a switch statement and nested loops to generate different decreasing patterns based on the user's choice, without using arrays. Each pattern is implemented using a separate parameterless function with no return value.

Algorithm

1. Start the program.
2. Display a menu with Decreasing Right-Angled Triangle, Decreasing Natural-Number Pattern, Decreasing Repeated-Number Pattern, Reverse-Number Triangle, and Exit.
3. Accept the user's choice.
4. Use a switch statement to call the corresponding function.
5. In the selected function, input the number of rows n.
6. Use nested loops to generate and display the required decreasing pattern.
7. After displaying the pattern, return to the main menu.
8. If the user enters an invalid menu choice, display an appropriate error message.
9. Continue displaying the menu until the user selects Exit.
10. Stop the program.

C Program

```
#include <stdio.h>

void starTriangle(void);
void naturalNumberPattern(void);
void repeatedNumberPattern(void);
void reverseNumberTriangle(void);

int main()
{
    int choice;

    do
    {
        printf("\n---- DECREASING PATTERNS MENU ----\n");
        printf("1. Decreasing Right-Angled Triangle\n");
        printf("2. Decreasing Natural-Number Pattern\n");
        printf("3. Decreasing Repeated-Number Pattern\n");
        printf("4. Reverse-Number Triangle\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: starTriangle(); break;
            case 2: naturalNumberPattern(); break;
            case 3: repeatedNumberPattern(); break;
```

```

        case 4: reverseNumberTriangle(); break;
        case 5: printf("Exiting program...\n"); break;
        default: printf("Invalid choice! Please enter a number from 1 to 5.\n");
    }
} while (choice != 5);
return 0;
}

void starTriangle(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}

void naturalNumberPattern(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}

void repeatedNumberPattern(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++) printf("%d ", i);
        printf("\n");
    }
}

void reverseNumberTriangle(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = i; j >= 1; j--) printf("%d ", j);
        printf("\n");
    }
}

```

Compilation/Execution Evidence

```
(kali㉿kali)-[~/Desktop/demo1]
$ gcc HDOC25.c
File System
(kali㉿kali)-[~/Desktop/demo1]
$ ./a.out
```

```
——— DECREASING PATTERNS MENU ———
1. Decreasing Right-Angled Triangle
2. Decreasing Natural-Number Pattern
3. Decreasing Repeated-Number Pattern
4. Reverse-Number Triangle
5. Exit
Enter your choice: 1
Enter number of rows: 4
* * * *
* * *
* *
*
*

——— DECREASING PATTERNS MENU ———
1. Decreasing Right-Angled Triangle
2. Decreasing Natural-Number Pattern
3. Decreasing Repeated-Number Pattern
4. Reverse-Number Triangle
5. Exit
Enter your choice: 2
Enter number of rows: 4
1 2 3 4
1 2 3
1 2
1
```

```

—— DECREASING PATTERNS MENU ——
1. Decreasing Right-Angled Triangle
2. Decreasing Natural-Number Pattern
3. Decreasing Repeated-Number Pattern
4. Reverse-Number Triangle
5. Exit
Enter your choice: 3
Enter number of rows: 4
4 4 4 4
3 3 3
2 2
1

```

```

—— DECREASING PATTERNS MENU ——
1. Decreasing Right-Angled Triangle
2. Decreasing Natural-Number Pattern
3. Decreasing Repeated-Number Pattern
4. Reverse-Number Triangle
5. Exit
Enter your choice: 4
Enter number of rows: 4
4 3 2 1
3 2 1
2 1
1

```

```

—— DECREASING PATTERNS MENU ——
1. Decreasing Right-Angled Triangle
2. Decreasing Natural-Number Pattern
3. Decreasing Repeated-Number Pattern
4. Reverse-Number Triangle
5. Exit
Enter your choice: 6
Invalid choice! Please enter a number from 1 to 5.

```

```

—— DECREASING PATTERNS MENU ——
1. Decreasing Right-Angled Triangle
2. Decreasing Natural-Number Pattern
3. Decreasing Repeated-Number Pattern
4. Reverse-Number Triangle
5. Exit
Enter your choice: 5
Exiting program ...

```

Test Case Table

Test Case	Choice	Input	Expected Output
1	1	n = 4	**** -> *** -> ** -> *
2	2	n = 4	1 2 3 4 -> 1 2 3 -> 1 2 -> 1
3	3	n = 4	4 4 4 4 -> 3 3 3 -> 2 2 -> 1
4	4	n = 4	4 3 2 1 -> 3 2 1 -> 2 1 -> 1
5	6	-	Invalid choice! Please enter a number from 1 to 5.
6	5	-	Program exits

Result

The menu-driven C program was successfully implemented using switch-case statements, parameterless functions, and nested loops without arrays. It correctly generates the required decreasing right-angled triangle, decreasing natural-number pattern, decreasing repeated-number pattern, and reverse-number triangle. The program continues displaying the menu until the Exit option is selected and also handles invalid menu choices.